

Engine

Discrete-event simulation in Julia: machine_shop (8 replications of 4000 minutes, seed 20260101)

Title: DiscreteEventSimulation | Model: machine_shop | Seed: 20260101 | Replications: 8 | Horizon: 4000 | Warmup: 400.0 |
Generated: | Engine: DiscreteSim | Sections: 8

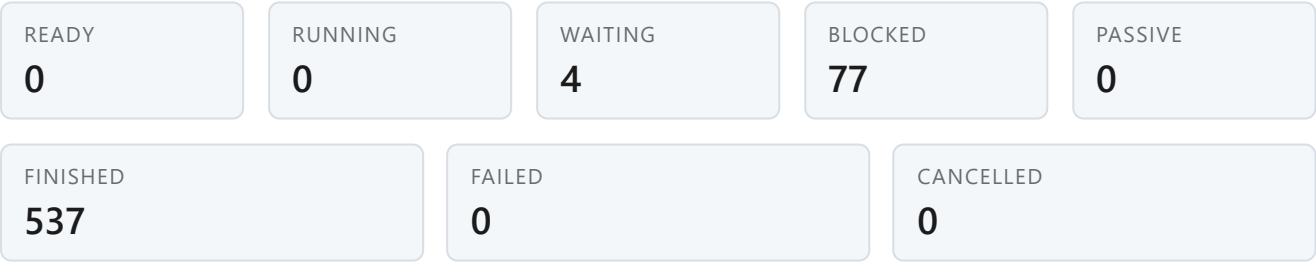
The engine

A next-event calendar, coroutines that suspend on two channels, and statistics that know whether they are adequate.

The run

Model machine_shop
Time unit minutes
Seed 3690686459077356
Events 7631
Clock 3999
Processes 618
Statistics 13
Streams 5
Trace rows 6223
Trace dropped 0
Stop reason horizon

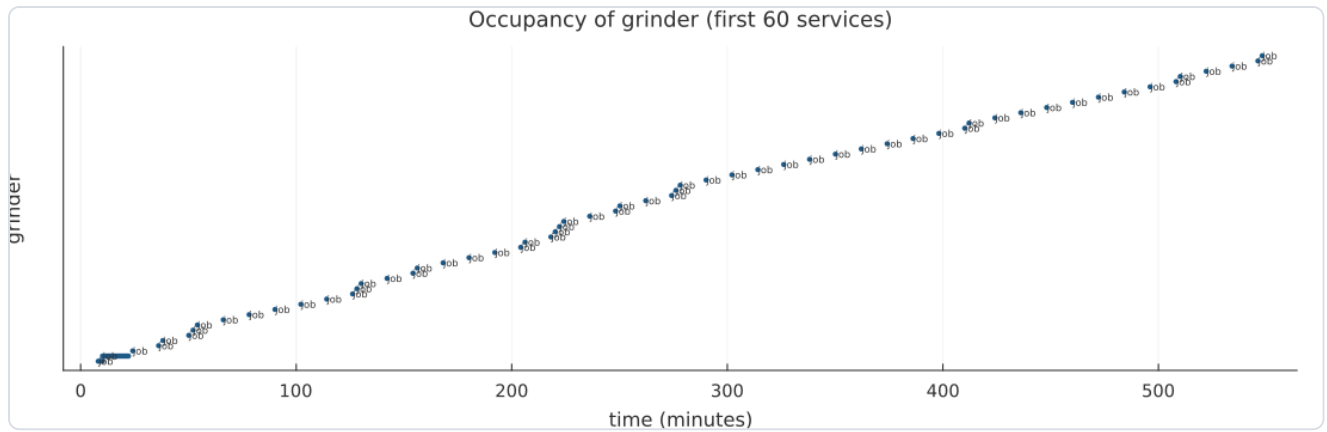
Where the processes ended up



Little's law, checked on every resource

Little law

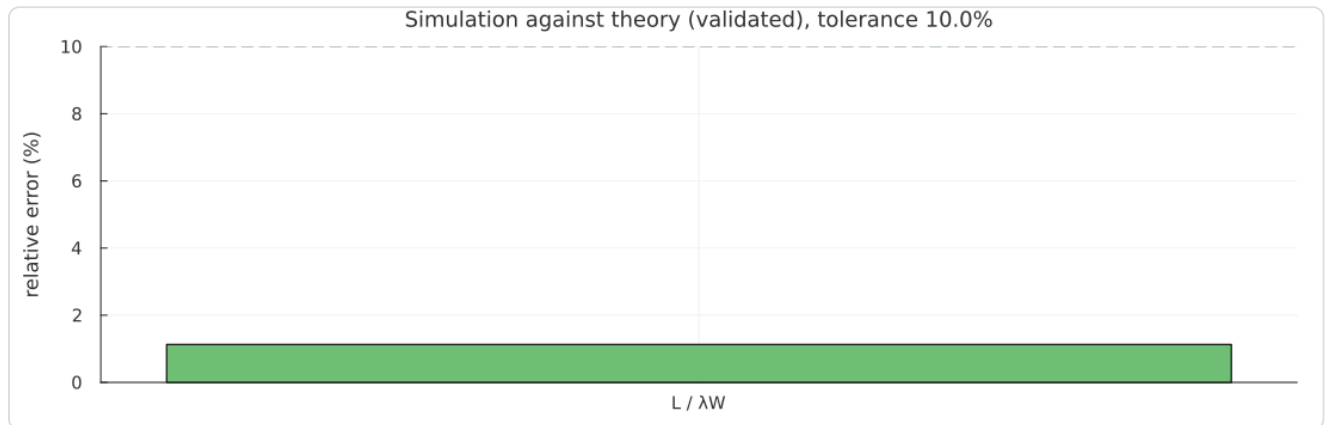
Resource	L	Λ	W	ΛW	Relative error	Holds
mill	6.578	0.092	75.6	6.971	0.06	no
drill	34.6	0.122	286.0	35.0	0.011	yes
grinder	36.1	0.103	325.4	33.5	0.07	no
lathe	1.535	0.083	18.5	1.537	0.00121	yes
press	1.069	0.079	13.4	1.066	0.00208	yes



Occupancy

Against theory

Verdict: **validated** worst relative error 1.13% against a tolerance of 10.0%.



Validation

Cross checks

Observed key	Theory key	Observed	Theory	Relative error	Within tolerance
L	λW	34.6	35.0	0.011	yes